

UNIT 5: INPUT/OUTPUT & FILE MANAGEMENT

Course Pattern: TE IT - 2024 Pattern (SPPU)
Status: Publication-Grade Theory Guide

Academic Term: Semester V (2026)
Course Code: PCC-301-ITT

Syllabus Mapping & Unit Structure

The following table outlines the curriculum mapping and priority weightage for Unit 5.

Syllabus Topic	In-Depth Exam Coverage	Priority / Weightage
I/O System Architecture	I/O Modes (Programmed I/O, Interrupt-Driven I/O, Direct Memory Access DMA), I/O Buffering strategies (Single, Double, Circular).	High Priority (6-8 Marks Qs)
Disk Architecture & Access	Hard disk geometry (Platters, Tracks, Sectors, Cylinders), Disk Access Time components (T_s , T_r , T_t).	High Priority (6-8 Marks Qs)
Disk Scheduling Algorithms	FCFS, SSTF, SCAN (Elevator), C-SCAN, LOOK, C-LOOK algorithms, Total head movement calculations.	High Priority (10-12 Marks Qs)
File System Structure	File Concept, File Attributes, Operations, Directory structures (Single-level, Two-level, Tree, Acyclic Graph).	Medium Priority (5-6 Marks Qs)
File Allocation & Unix Inodes	Allocation methods (Contiguous, Linked FAT, Indexed), Unix Inode Structure (Direct/Indirect blocks), Free space management, Security.	High Priority (10-12 Marks Qs)

Input/Output System Architecture and Buffering

Organization of I/O Execution Functions

- Programmed I/O:**
 - The CPU directly controls I/O operations by continually polling status registers of the I/O module until the operation completes.
 - Drawback:** Severe CPU busy-waiting (spinlock), wasting valuable processing cycles.
- Interrupt-Driven I/O:**
 - The CPU issues an I/O command to the device module and immediately switches execution to another process.
 - When the device completes I/O, it raises a hardware **Interrupt Signal** on the interrupt bus, forcing the CPU to execute the corresponding Interrupt Service Routine (ISR).
- Direct Memory Access (DMA):**
 - Designed for high-speed bulk data transfers (HDDs, SSDs, Network Interface Cards).
 - A dedicated **DMA Controller** transfers entire data blocks directly between physical RAM memory and the I/O module without CPU intervention.
 - The CPU is interrupted only twice: once at transfer initialization and once when the entire block transfer finishes.

I/O Buffering Strategies

1. **Single Buffering:** OS assigns a single memory buffer in kernel RAM. While the user process consumes data from the buffer, the I/O device transfers the next block into kernel space.
2. **Double Buffering (Ping-Ponging):** OS assigns two kernel memory buffers (B_1 and B_2). The device fills B_1 while the process reads B_2 , then swaps roles seamlessly.
3. **Circular Buffering:** OS maintains a queue of three or more buffers managed via producer-consumer semantics to handle bursty I/O throughput.

Parameter	Interrupt-Driven I/O	Direct Memory Access (DMA)
CPU Involvement	High (CPU handles every byte/word transfer via ISR).	Minimal (CPU initializes block count & base address).
Transfer Mechanism	Data moves through CPU registers (<i>Dto CPU to RAM</i>).	Data moves directly over system bus (<i>Dto RAM</i>).
Interrupt Frequency	Interrupts CPU after every single byte or word.	Interrupts CPU once per entire data block transfer.
Optimal Use Case	Low-speed byte devices (Keyboard, Mouse, Serial ports).	High-speed block devices (Disk drives, 10GbE NICs).

Source: Stallings, Ch 11, p. 440-465 | Source: Silberschatz, Ch 13, p. 515-540

Hard Disk Architecture and Disk Scheduling

Hard Disk Geometry and Access Time Math

A magnetic Hard Disk Drive (HDD) consists of stacked physical **Platters** spinning on a central spindle (5400 – 15000 RPM).

- Each platter surface is coated with magnetic material divided into concentric rings called **Tracks**.
- Identical track positions across all platter surfaces form a **Cylinder**.
- Tracks are subdivided into 512-byte or 4096-byte **Sectors**.

Components of Disk Access Time

1. **Seek Time (T_s):** Time required for the read/write head arm to move physically to the target cylinder track.
2. **Rotational Latency (T_r):** Time required for the target sector to rotate underneath the read/write head.
 - Average Rotational Latency: $T_{\{r,avg\}} = \frac{1}{2 * \text{RPM}}$ minutes.
3. **Transfer Time (T_t):** Time required to transfer the requested data bytes.
 - $T_t = \frac{b}{r * N}$, where b is bytes to transfer, r is rotation speed, and N is bytes per track.

$$\text{Total Disk Access Time} = T_s \text{ (Seek Time)} + T_r \text{ (Rotational Latency)} + T_t \text{ (Transfer Time)}$$

Seek Time (3-10 ms) dominates total access time $\Rightarrow$ Disk Scheduling minimizes Seek Head Movement!

Figure 5.1: Components of Total Hard Disk Access Time

Disk Scheduling Algorithms Comparison

Algorithm	Head Movement Rule	Key Advantages	Major Drawbacks
FCFS	Serves requests in exact order of arrival.	Completely fair; zero starvation risk.	Wild head swings; very high total seek distance.
SSTF	Serves request closest to current head position.	Minimizes average seek time.	Starvation of requests located far from active head area.
SCAN (Elevator)	Moves head toward one end, servicing requests until disk boundary, then reverses.	Low variance in response time; eliminates starvation.	Favors requests near boundary turnaround points.
C-SCAN	Moves head in one direction servicing requests; jumps back to start without servicing.	Provides uniform wait time across all cylinders.	Extra unserved return trip overhead.
LOOK / C-LOOK	Same as SCAN/C-SCAN, but head reverses immediately at last request (no boundary trip).	Prevents unnecessary head travel to physical disk ends.	Preferred standard for production disk controllers.

Source: Stallings, Ch 11, p. 475-500 | Source: Silberschatz, Ch 10, p. 465-490

File Systems, Allocation Methods, and Unix Inodes

File Allocation Methods

1. Contiguous File Allocation:

- Each file occupies a contiguous set of disk blocks.
- Advantage:** Maximum sequential read performance (single seek operation).
- Disadvantage:** External fragmentation and difficulty expanding file size.

2. Linked File Allocation (FAT):

- Each file is a linked list of disk blocks; directory entry contains pointer to first and last blocks.
- Advantage:** Zero external fragmentation; easy file expansion.
- Disadvantage:** Slow random access (requires traversing links sequentially); pointer corruption destroys file.

3. Indexed File Allocation (Unix Inode):

- Each file possesses a dedicated index block (**Inode**) containing an array of direct and indirect disk block pointers.
- Advantage:** Fast direct and random access; supports dynamic file expansion without external fragmentation.

Inode Block: Mode, Owner, Size, Timestamps

Direct Block Pointers (0 to 11) → Direct Data Blocks

Single Indirect Pointer → Block of Direct Pointers → Data Blocks

Double Indirect Pointer → Block of Single Indirect Pointers → Data Blocks

Triple Indirect Pointer → Block of Double Indirect Pointers → Data Blocks

Figure 5.2: Structure of Unix Inode with Direct & Multi-Level Indirect Pointers

Directory Structures and Protection

1. **Directory Structures:** Single-Level, Two-Level, Tree-Structured (Paths: /usr/home/file.txt), Acyclic Graph (Shared files via hard/soft links).
2. **File Protection & Access Control:**
 - **Access Control Matrix:** Grid specifying explicit permissions for each User U_i on File F_j .
 - **Access Control Lists (ACL):** Attached to each file, listing users and permitted operations (rwx).
 - **Capability Lists:** Attached to each user, listing accessible files and rights.

Source: Stallings, Ch 12, p. 510-545 | Source: Tanenbaum, Ch 4, p. 260-295

End of Operating Systems Unit 5 Theory Notes — SPPU TE IT 2024 Pattern